No. of Functional Features Included in the Solution

Date	27 June 2026
Team ID	SWTID-2026-5245
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done/In Progress/Pending)	Marks Contribution
1	Soil & Climate Input Form	Web form where the farmer enters N, P, K, temperature, humidity, pH and rainfall, with client-side min/max validation.	Frontend (templates/index.html)	Done	High
2	Crop Recommendation Engine	Trained scikit-learn classifier predicts the most suitable crop from the submitted soil and climate values.	Backend (api/index.py, model/crop_model.pkl)	Done	High
3	Confidence Score Display	Shows the model's prediction confidence (%) alongside the recommended crop on the result page.	Backend + Frontend (result.html)	Done	Medium
4	Client-Side Input Validation	HTML min/max/step constraints on each input field to reduce invalid submissions before they reach the server.	Frontend (templates/index.html)	Done	Medium
5	Web Deployment	Application packaged and deployed as a serverless Flask app on Vercel, accessible via a public URL.	Deployment (vercel.json, Gunicorn)	Done	High
6	Crop Suitability Check	Allow a farmer to check whether a specific crop they already have in mind is suitable for their current soil/climate (Scenario 2).	Backend / Frontend (not yet built)	Pending	Medium
7	Researcher/Policymaker Analytics Dashboard	Aggregated view of crop-environment data and trends for researchers and policymakers (Scenario 3).	New module (not yet built)	Pending	Medium
8	Weather API Integration & Resource Tips	Auto-fetch real-time weather data and show fertilizer/resource-efficiency tips alongside the recommendation.	External API integration (not yet built)	Pending	Low

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	5
Core / Must-Have Features	4
Additional / Nice-to-Have Features	4
Features Tested & Verified	5

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Soil & Climate Input Form; Recommendation Result Display
2	Backend / Logic	2	Crop Recommendation Engine (ML inference); Input parsing & prediction route
3	Database / Storage	0	None currently – model loaded from serialized .pkl files, no live database
4	API / Integration	1 (planned)	Weather Data API (planned, not yet integrated)
5	Security / Authentication	0	None currently – application is openly accessible without login